CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

Perceived security of trips on foot by gender (SOC_SC_PS2)

DOMAIN

Transport	Environment	Energy	**Society**	Economy

TOPIC	**Security**	
IMPACT	**Perceived security of trips on foot** *Increasing the perception that walking in the pilot area is safe*	**SOC_SC**

Category

Key indicator	**Supplementary indicator**	State indicator

CONTEXT AND RELEVANCE

Perceived travel security refers to how safe individuals feel when making trips within the pilot area. This indicator measures perceived safety while travelling on foot. It accounts for concerns such as the risk of assault, harassment, and sexual harassment in public spaces. Perceptions of safety are shaped by a range of factors, including urban design, lighting, maintenance and cleanliness, as well as the presence and behaviour of others. They reflect individuals' emotional responses to the environment and are influenced by past personal experiences and media reporting of incidents.

Perceived security can vary across population groups. In particular, women and gender-diverse individuals may face higher exposure to harassment or unsafe situations, which can affect their mobility choices. Higher perceived safety when walking can increase willingness to walk, reduce reliance on private vehicles, promote social inclusion, and support more sustainable urban mobility patterns.

This is a relevant indicator when the policy action is aimed at increasing the perceived security of trips on foot. A successful action is reflected in a HIGHER value of the indicator.

DESCRIPTION

This indicator is a vector of **dimensionless scores**, each representing the average perceived level of security of trips on foot for a given gender, as reported by a sample of individuals in the pilot area.

METHOD OF CALCULATION AND INPUTS

The indicator is calculated as the average score by gender assigned by a sample of citizens who provided responses to a question asking how secure they feel making trips on foot in the pilot area. **The indicator should be calculated by the user** based on the inputs described below and its values should be entered in the supporting tool.

Method	
Calculation of the indicator based on responses collected by means of a sample survey	Significance: **0.50**

INPUTS

The following information is needed to compute the indicator:

a) **Responses of a sample of individuals to questions i) regarding their perceived security when travelling on foot in the pilot area, and ii) regarding their gender.**

A suggested formulation of the questions is provided in the Guidelines for surveys which are part of the MUSE Evaluation Framework. It is recommended that the question asks the respondent to rate their perceived security when travelling on foot **alone**.

The experiment would be reflected in terms of a different responses to the same question on perceived security.

METHOD OF CALCULATION

The requirement for computing the indicator is collecting the responses from a sample survey (which can be organised to collect more information than the one needed for this indicator). Assuming that the formulation of the questions suggested in the Guidelines for surveys is used, the indicator should be computed **by the user** according to the following steps:

- **Calculation of the share of individuals assigning a certain score to the security of trips on foot by gender** (see equation below)
- **Calculation of the perceived security indicator by gender** (see equation below).

EQUATIONS

The share of individuals of a certain gender assigning a certain score to the security of trips on foot should be computed as:

$$^{l}FTSecSh_g = \frac{\sum_i (^{s}FTSec_{i,g}\ where\ s = l)}{I_g}$$

Where:

$^{s}FTSec_{i,g}$ = Score of travel security s assigned by individual i of gender g.

I_g = Total number of responses collected for gender g

g = Gender category

l = Score level (numerical score 1–10).

For example, in a sample of 600 individuals, if 250 of these were men and 134 of them assigned a travel security score of 6, $\sum_i {}^{s}FTSec_{i,g}\ where\ s = 6$ would be 134 and the share of men assigning this score would be 134 / 250 = 54%

The indicator representing the perceived security of trips on foot is a vector, with each value corresponding to a specific gender. It should be calculated as follows:

$$\boldsymbol{AvPercFTSec_g = \sum_l \left(^{l}FTSecSh_g * {}^{l}FTSecScore\right)}$$

Where:

$^{l}FTSecScore$ = Perceived security score *l*.

In the example above, the travel security score *l* is 6.

ALTERNATIVE INDICATORS

This indicator refers to the perceived security of trips on foot. Indicator **SOC_SC_PS1** measures instead the perceived security of trips on public transport.

Furthermore, alternative indicators in the CIVITAS Evaluation Framework quantify of objective levels of public transport security. They are **SOC_SC_SC1** and **SOC_SC_SC2**. The difference between one or the other lies mainly in the type and source of data used for computation. SOC_SC_SC1 assesses transport-related security by measuring the share of residents in the pilot area who have experienced sexual harassment on public transport or at stops and stations. Data are collected

through a sample survey. SOC_SC_SC2, on the other hand, tracks the number of crime incidents occurring on board public transport and at stops or stations, based on police records. Since it relies on official data, SOC_SC_SC2 is the most significant and generally easier to compute, as data collection only consists of retrieving police statistics. However, SOC_SC_SC1 may be preferable when the aim is to specifically assess the prevalence of sexual harassment and to capture unreported incidents which are missing from official records.